PRASAD CHAVAN

+91 9373719870 | prasad2004.ac@gmail.com| Pune, Maharashtra

EDUCATION

Pune Vidyarthi Griha's College of Engineering and Technology, Pune Nov 2022 - Present
Bachelor of Engineering in Mechanical **GPA: 7.80/10**

G. K. Sapre Junior College Of Science, Devrukh, Ratnagiri 2022
Maharashtra State Board of Secondary and Higher Secondary Education, HSC **Score : 72.50%**

New English School, Devrukh, , Ratnagiri 2020
Maharashtra State Board of Secondary and Higher Secondary Education, SSC **Score : 89.00%**

EXPERIENCE

R&D Design Intern – Aeron Systems, Pune **Feb 2026 – Jul 2026**

- Created SolidWorks assemblies, detailed drawings, and step-by-step assembly instruction drawings
- Prepared BOM, MICD, and engineering documentation per internal standards
- Applied GD&T in detailed part drawings and managed assembly configurations
- Developed VBA macros in SolidWorks to automate tasks like figure label renumbering across multi-sheet drawings
- Designed a prototype robot for automated cleaning of pyranometer and albedometer lenses used in weather monitoring systems

Intern – SGM Tech, Pune | **Jan 2025 – Feb 2025**

- Participated in the complete design and development of a boiler manufacturing plant, from layout planning to system integration.
- Assisted in mechanical design and detailing of boiler components, piping layouts, and auxiliary systems using CAD tools.
- Worked on plant layout, material flow planning, and equipment placement to optimize manufacturing efficiency and safety.

Vehicle Dynamics Engineer – Team Conquistador, PVG's COET | **Feb 2023**

- Designed ATV suspension system for Baja SAE; performed modelling in SolidWorks and analysis in ANSYS
- Achieved 1st Runner-Up in Suspension & Traction at ATVC competition

PROJECTS

Power Generation from Train Track Jun 2023

- Designed a system to harness train motion for electricity generation using a **piston-based mechanism**
- Converted kinetic energy into electrical energy through a **crankshaft-driven generator**

B.E. Final Year Thesis: Acoustic Vehicle Alerting System (AVAS) for Pedestrian Safety Jun 2025
(Ongoing)

- Developing a regulatory-compliant Acoustic Vehicle Alerting System (AVAS) to actively mitigate pedestrian safety risks associated with quiet electric vehicle operation.
- Researching and implementing advanced psychoacoustics and signal processing to design an effective and context-aware audible warning signal.

SKILLS

- **CAD Software:** SolidWorks – 3D modelling, assembly, and detailed drawings
- **CAE Software:** ANSYS – structural, thermal, and basic CFD analysis
- **Additive Manufacturing:** 3D printing design and prototyping
- **Macros in solidWork**

RESEARCH EXPERIENCE

- Conducted research on **Acoustic Vehicle Alerting System (AVAS)** for pedestrian safety; authored and submitted a review paper to *Traffic Injury Prevention Journal (Q1)*.
- Developed a research-based **suspension system design**, focusing on optimization of suspension geometry, load distribution, and ride–handling balance through CAD modelling and simulation for performance-oriented vehicle applications.